Postulates, Theorems, Converses, and Biconditionals Answer Key

High-school geometry

Quick Answers

1. —
2. 6
3. —
4. (a) x = 13, (b) the measure of the smaller part = 21 degrees
5. —
6. —
7. 22
8. —
9. 20
10. (b) x = 12, —

What is verified

4 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 3, 5, 6, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry
HSG-CO.A – *problems 1, 2, 4, 5, 8, 10*
HSG-CO.C – *problems 3, 6, 7, 9*
Difficulty 1–5 of 5 across 16 tagged checks.

1. Label four statements P (postulate) or T (theorem), and name the test used.

1. P — "through any two points there is exactly one line" is one of the starting assumptions; nothing earlier could prove it. **2. T** — vertical angles are congruent is proved from the Linear Pair Postulate and subtraction. **3. P** — another starting assumption about points, lines and planes. **4. T** — proved from the definition of supplementary and subtraction.

The test: ask whether the statement is *accepted* or *earned*. A postulate is where the system starts; a theorem has a proof behind it that traces back to postulates and definitions. If you can imagine writing a two-column proof of it, it is a theorem.

Grading: four correct labels and a test stated in terms of proof, not in terms of how obvious the statement feels.

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2. Segment Addition: $AB = 2x + 3$, $BC = x - 1$, $AC = 20$. Find x.

Solution: the postulate turns the picture into one equation.

$$\begin{aligned}(2x+3)+(x-1)&=20\\ 3x+2&=20\\ 3x&=18\end{aligned}$$

So $x = 6$. Check: $AB = 15$, $BC = 5$, and $15 + 5 = 20$. $\boxed{x = 6}$

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3. Vertical angles theorem: (a) converse, (b) is the converse true?

(a) Swap hypothesis and conclusion: *"If two angles are congruent, then they are vertical angles."*

(b) False. Counterexample: draw two separate 50° angles anywhere on the page. They are congruent, so the converse's hypothesis holds, but they are not vertical angles — vertical angles are the *opposite pair formed by two intersecting lines*, and these two share no vertex at all. The base angles of an isosceles triangle work equally well.

Grading: (a) must swap the two halves exactly. (b) needs a congruent pair plus the reason they fail the definition of vertical angles.

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4. Angle Addition: $m\angle ABD = 3x + 5$, $m\angle DBC = 2x - 5$, $m\angle ABC = 65$. (a) x. (b) $m\angle DBC$.

(a) The postulate gives one equation:

$$\begin{aligned}(3x+5)+(2x-5)&=65\\ 5x&=65\end{aligned}$$

So $x = 13$. $\boxed{x = 13}$

(b) $m\angle DBC = 2(13)-5 = 26-5$. (Check: $m\angle ABD = 3(13)+5 = 44$, and $44+21 = 65$.)

$\boxed{m\angle DBC = 21^\circ}$

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5. "An angle is a right angle if and only if its measure is 90°." (a) The two conditionals. (b) Why both are needed.

(a) Forward: *if an angle is a right angle, then its measure is* 90°. Converse: *if an angle measures* 90°*, then it is a right angle.*

(b) "If and only if" claims both directions at once, so the biconditional is true exactly when both conditionals are true. If only the forward direction held, the phrase would still promise you could go backwards — and a statement that promises more than it can deliver cannot be used as a reason in a proof. That is why definitions are written as biconditionals and most theorems are not.

Grading: (a) both directions, correctly worded. (b) must connect "if and only if" to the truth of *both* conditionals.

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6. Linear Pair Postulate: (a) converse, (b) is it true?

(a) *"If two angles are supplementary, then they form a linear pair."*

(b) False. A 110° angle drawn at the top of the page and a 70° angle drawn at the bottom are supplementary — $110+70=180$ — but they are not a linear pair. A linear pair must be adjacent and its two non-common sides must be opposite rays; these angles share neither a vertex nor a side.

Grading: a supplementary pair that is not adjacent, plus the adjacency or opposite-ray condition named.

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7. Linear pair with measures $5x-8$ and $3x+12$. Find x.

Solution: the postulate says the two measures are supplementary, so they add to 180.

$$\begin{aligned}(5x-8)+(3x+12)&=180\\ 8x+4&=180\\ 8x&=176\end{aligned}$$

So $x=22$. Check: $5(22)-8=102$ and $3(22)+12=78$, and $102+78=180$. $\boxed{x=22}$

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8. Congruent segments. (a) Write the biconditional. (b) Name a statement on this sheet that cannot be made biconditional.

(a) *"Two segments are congruent if and only if they have the same length."* You are entitled to write it because *both* directions hold: congruent segments have equal lengths, and segments of equal length are congruent. That two-way truth is what makes this a definition rather than a theorem.

(b) Either the vertical-angle theorem or the Linear Pair Postulate. Take the linear pair one: forwards it is true, but its converse (problem 6) is false, so joining them with "if and only if" would assert that every supplementary pair is a linear pair — which the 110°/70° counterexample refutes.

Grading: (b) needs a statement whose converse was shown false *and* that reason stated.

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9. Converse of Corresponding Angles: $(4x+15)°$ and $(6x-25)°$. Find x making $j \parallel k$.

Solution: the converse lets you *conclude* parallel lines from congruent corresponding angles, so set the two measures equal.

$$\begin{aligned}4x+15&=6x-25\\ 40&=2x\end{aligned}$$

So $x=20$. Check: both angles measure $4(20)+15=95$ and $6(20)-25=95$. $\boxed{x=20}$

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10. "Two lines are perpendicular if and only if they form a right angle." (a) The two conditionals and their uses. (b) Find x for $(7x+6)°$.

(a) Forward: *if two lines are perpendicular, then they form a right angle.* Converse: *if two lines form a right angle, then they are perpendicular.* Given perpendicular lines, you use the forward direction to obtain a 90° angle you can compute with. To *prove* lines perpendicular, you first show some angle measures 90° and then use the converse to draw the conclusion. Same statement, opposite directions of travel.

(b) Using the converse, the angle must be a right angle:

$$\begin{aligned} 7x + 6 &= 90 \\ 7x &= 84 \end{aligned}$$

So $x = 12$. $\boxed{x = 12}$

Grading: (a) both conditionals plus the given-versus-concluded distinction.